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🎯 Quiz: Model Fine-Tuning & LLM Optimization

🧠 Section 1: Conceptual MCQs

1. What is model fine-tuning?
A. Training a model from scratch
B. Adjusting a pre-trained model on new data
C. Only changing prompts
D. Increasing model size

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2. Which scenario is best suited for prompting instead of fine-tuning?
A. Domain-specific chatbot
B. Repetitive structured output task
C. One-time query with general knowledge
D. Training on private dataset

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3. What is the main advantage of fine-tuning over prompting?
A. Faster inference always
B. Better control over task-specific outputs
C. No need for data
D. Lower memory usage

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4. What does LoRA stand for?
A. Low Rank Adaptation
B. Linear Representation Adjustment

C. Layer Optimization and Reuse Algorithm
D. Learning Rate Adjustment

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5. In LoRA, what is actually trained?
A. Entire model weights
B. Only input tokens
C. Small low-rank matrices
D. Output layer only

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6. Why is LoRA considered parameter-efficient?
A. It reduces dataset size
B. It updates only a small subset of parameters
C. It removes layers
D. It compresses input

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7. What is the role of a tokenizer in fine-tuning?
A. Train model weights
B. Convert text into tokens
C. Evaluate model
D. Store embeddings

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8. Which step comes first in a fine-tuning workflow?
A. Model evaluation
B. Data collection & cleaning
C. Training
D. Deployment

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9. What is overfitting in fine-tuning?
A. Model performs well on all data
B. Model memorizes training data but fails on new data
C. Model ignores training data
D. Model becomes faster

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10. Which metric is commonly used for classification tasks?
A. BLEU score
B. Accuracy
C. Perplexity
D. Latency

🔍 Section 2: True / False

11. Fine-tuning always requires large datasets. → False
12. Prompting does not change model weights. → True
13. LoRA modifies all layers of the model. → False
14. Validation data is used to tune hyperparameters. → True
15. Fine-tuned models can outperform base models on niche tasks. → True

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🧩 Section 3: Scenario-Based Questions

16. Scenario:
You need a chatbot specialized in legal documents with consistent tone and formatting.
👉 What should you choose?
A. Prompting
B. Fine-tuning
C. Zero-shot learning
D. Random initialization

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17. Scenario:
You want quick results for generating marketing slogans without training.
👉 Best approach?
A. Fine-tuning
B. Prompt engineering
C. LoRA
D. Training from scratch

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18. Scenario:
You have limited GPU resources but want to adapt a large model.
👉 Best method?
A. Full fine-tuning
B. LoRA
C. Increase batch size
D. Add more layers

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19. Scenario:
Your model performs well on training data but poorly on test data.
👉 What is the issue?
A. Underfitting
B. Overfitting
C. Data leakage
D. Tokenization error

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20. Scenario:
You are comparing two models for summarization.
👉 Which metric is most relevant?
A. Accuracy
B. BLEU / ROUGE
C. Latency only
D. Token count

⚙️ Section 4: Short Answer

21. Explain one difference between prompting and fine-tuning.

Ans : Prompting gives instructions to the model at runtime without changing it, while fine-tuning modifies the model's weights by training on new data.

22. What is one benefit of using LoRA?

Ans: It reduces the number of trainable parameters, making training faster and more memory-efficient.

23. List any two steps in a fine-tuning workflow.

Ans:

1. Prepare the Dataset

2. Train (Fine-Tune) the Model

24. What is the purpose of validation data?

Ans: Validation data is used to evaluate the model during training and check how well it generalizes to unseen data.

25. Name one metric used to evaluate LLMs.

Ans: Perplexity

🧪 Section 5: Practical Thinking

26. You are fine-tuning a model and loss is not decreasing. What could be one reason?

Ans: Learning rate is too high

27. Why might fine-tuning be expensive?

Ans: Fine-tuning can be expensive because it requires significant computational resources (like GPUs) to train the model.

28. When should you avoid fine-tuning?

Ans: You should avoid fine-tuning when you don't have enough labeled data.

29. What happens if tokenizer is mismatched with the model?

Ans: The input text will be incorrectly processed, leading to poor or incorrect predictions.

30. How does LoRA reduce computational cost?

Ans: LoRA reduces computational cost by updating only a small number of additional (low-rank) parameters instead of the entire model.